Linear Algebra and Its Applications

Section 6.7: Inner Product Spaces

Detailed worked solutions

Mathjock

Text and numbering. David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

Source scope. The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

How to read the solutions. Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

Contents

6.7 Inner product spaces 3

Practice problem 6.7.P1: A zero vector has zero inner product with every vector 3

Practice problem 6.7.P2: Additivity in the second argument 3

Exercise 6.7.1: A weighted norm and its orthogonal line 3

Exercise 6.7.2: Verify Cauchy–Schwarz with the weighted product 3

Exercise 6.7.3: Evaluate before multiplying 4

Exercise 6.7.4: A negative inner product is allowed 4

Exercise 6.7.5: Norms from the first pair of value lists 4

Exercise 6.7.6: Norms from the second pair 4

Exercise 6.7.7: Project onto the first polynomial line 4

Exercise 6.7.8: Project onto the second polynomial line 4

Exercise 6.7.9: Remove the lower-degree part of a quadratic 5

Exercise 6.7.10: Best quadratic approximation to an odd cubic 5
Exercise 6.7.11: A cubic projected onto a five-node quadratic space 5
Exercise 6.7.12: Choose a scaled orthogonal cubic 5
Exercise 6.7.13: Transport the Euclidean product through an invertible map 6
Exercise 6.7.14: The same construction for an injective transformation 6
Exercise 6.7.15: Move a scalar out of the second argument 6
Exercise 6.7.16: The distance between two orthonormal vectors 6
Exercise 6.7.17: Recover an inner product from two norms 6
Exercise 6.7.18: The parallelogram identity 6
Exercise 6.7.19: Arithmetic mean dominates geometric mean 7
Exercise 6.7.20: The square of a mean versus the mean of squares 7
Exercise 6.7.21: An integral inner product that vanishes 7
Exercise 6.7.22: Another orthogonal pair of functions 7
Exercise 6.7.23: An integral norm is a square root 7
Exercise 6.7.24: The norm of a cubic-minus-quadratic function 7
Exercise 6.7.25: An orthogonal polynomial basis on a symmetric interval 8
Exercise 6.7.26: Repeat the orthogonalization on a wider interval 8
Exercise 6.7.27: Complete the five-node orthogonal polynomial basis 8
Exercise 6.7.28: Orthogonalize powers of cosine by integration 8

6.7 Inner product spaces

An inner product assigns real numbers $\langle u, v\rangle$ and satisfies symmetry, additivity, homogeneity, and positive definiteness. The associated norm is $\|u\| = \sqrt{\langle u, u\rangle}$; orthogonality means $\langle u, v\rangle = 0$. The formula depends on the space: a weighted coordinate product, evaluation of polynomials, and integration of functions are different inner products. For a nonzero vector p, projection onto its span is $\langle q, p\rangle p/\langle p, p\rangle$. For an orthogonal list, add these separate projections. Always use the inner product stipulated in the exercise rather than substituting the ordinary coordinate dot product.

Practice problem 6.7.P1: A zero vector has zero inner product with every vector

By homogeneity, $\langle 0, v\rangle = \langle 0u, v\rangle = 0\langle u, v\rangle = 0$, for any u in the space. Symmetry then gives $\langle v, 0\rangle = \langle 0, v\rangle = 0$. The choice of u does not matter, because its scalar multiple is zero.

Practice problem 6.7.P2: Additivity in the second argument

Use symmetry to move the sum into the first argument, where additivity is an axiom:

$$\langle u, v + w\rangle = \langle v + w, u\rangle = \langle v, u\rangle + \langle w, u\rangle = \langle u, v\rangle + \langle u, w\rangle.$$

The last equality applies symmetry to each term separately.

Exercise 6.7.1: A weighted norm and its orthogonal line

Example 1 uses $\langle u, v\rangle = 4u_1v_1 + 5u_2v_2$.

(a) Apply those weights to the stated vectors.

For $x = (1, 1)$ and $y = (5, -1)$, $\|x\|^2 = 4 + 5 = 9$, $\|y\|^2 = 4(25) + 5 = 105$, and $\langle x, y\rangle = 20 - 5 = 15$. Thus $\|x\| = 3$, $\|y\| = \sqrt{105}$, and $|\langle x, y\rangle|^2 = 225$.

(b) Solve the orthogonality equation.

A vector $z = (z_1, z_2)$ is orthogonal to y exactly when $20z_1 - 5z_2 = 0$, or $z_2 = 4z_1$.

Conclusion. All such vectors are $t(1, 4)$, $t \in \mathbb{R}$, including zero.

Exercise 6.7.2: Verify Cauchy–Schwarz with the weighted product

For $x = (3, -2)$ and $y = (-2, 1)$,

$$\langle x, y\rangle = 4(3)(-2) + 5(-2)(1) = -34,$$

$$\|x\|^2 = 4(9) + 5(4) = 56, \qquad \|y\|^2 = 4(4) + 5(1) = 21.$$

Therefore $|\langle x, y\rangle|^2 = 1156 \le 1176 = \|x\|^2\|y\|^2$. Both sides of $|\langle x, y\rangle| \le \|x\|\|y\|$ are nonnegative, so the squared comparison proves the desired inequality.

Evaluation convention for Exercises 3–8. The inner product is the dot product of the three value lists at $-1, 0, 1$, not of the polynomial coefficient lists. For Exercise 3, p and q have value vectors $(3, 4, 5)$ and $(1, 5, 1)$. For Exercise 4, their value vectors are $(-4, 0, 2)$ and $(5, 3, 5)$.

Exercise 6.7.3: Evaluate before multiplying

For $p(t) = 4 + t$ and $q(t) = 5 - 4t^2$,

$$\langle p, q\rangle = p(-1)q(-1) + p(0)q(0) + p(1)q(1) = 3(1) + 4(5) + 5(1) = \boxed{28}.$$

Each product uses values at the same input.

Exercise 6.7.4: A negative inner product is allowed

For $p(t) = 3t - t^2$ and $q(t) = 3 + 2t^2$,

$$\langle p, q\rangle = (-4)(5) + 0(3) + 2(5) = \boxed{-10}.$$

Only self-products must be nonnegative; products of different vectors can be negative, as this example shows.

Exercise 6.7.5: Norms from the first pair of value lists

Using the value vectors $(3, 4, 5)$ and $(1, 5, 1)$,

$$\|p\| = \sqrt{9 + 16 + 25} = \boxed{5\sqrt{2}}, \qquad \|q\| = \sqrt{1 + 25 + 1} = \boxed{3\sqrt{3}}.$$

The norm is the square root of the sum of squared values, not its square.

Exercise 6.7.6: Norms from the second pair

The value vectors are $(-4, 0, 2)$ and $(5, 3, 5)$, so

$$\|p\| = \sqrt{16 + 0 + 4} = \boxed{2\sqrt{5}}, \qquad \|q\| = \sqrt{25 + 9 + 25} = \boxed{\sqrt{59}}.$$

These are norms in the specified evaluation inner product.

Exercise 6.7.7: Project onto the first polynomial line

For the pair in Exercise 3, $\langle q, p\rangle = 28$ and $\langle p, p\rangle = 50$. Since $p \neq 0$, projection onto $\mathrm{Span}\{p\}$ is

$$\mathrm{proj}_{\mathrm{Span}\{p\}}\, q = \frac{28}{50}p = \boxed{\frac{14}{25}(4 + t)}.$$

A check is $\langle q - (14/25)p, p\rangle = 28 - (14/25)50 = 0$. Thus the residual is orthogonal to the whole one-dimensional target space.

Exercise 6.7.8: Project onto the second polynomial line

For the pair in Exercise 4, the coefficient is $\langle q, p\rangle/\langle p, p\rangle = -10/20 = -1/2$. Hence the projection is $\boxed{-\frac{1}{2}(3t - t^2) = (t^2 - 3t)/2}$. Its residual check is $-10 - (-1/2)20 = 0$. A negative projection coefficient is perfectly possible.

Exercise 6.7.9: Remove the lower-degree part of a quadratic

Use evaluation nodes $-3, -1, 1, 3$. The value lists for p_0, p_1, p_2 are $(1, 1, 1, 1)$, $(-3, -1, 1, 3)$, and $(9, 1, 1, 9)$.

(a) Project onto the first two orthogonal polynomials.

The first two lists have dot product zero. Also $\langle p_2, p_0\rangle = 20$, $\langle p_0, p_0\rangle = 4$, and $\langle p_2, p_1\rangle = -27 - 1 + 1 + 27 = 0$. Thus the projection is $5p_0 = 5$.

(b) Form and scale the orthogonal remainder.

The remainder $t^2 - 5$ has values $(4, -4, -4, 4)$. Divide by four to get $\boxed{q(t) = (t^2 - 5)/4}$, whose values are $(1, -1, -1, 1)$. It is orthogonal to both p_0, p_1 and is nonzero. The three polynomials have distinct increasing degrees, so they are independent and span the same space as $1, t, t^2$.

Exercise 6.7.10: Best quadratic approximation to an odd cubic

In the same four-node inner product, t^3 is orthogonal to the even polynomials p_0 and q: the products cancel at opposite nodes. Its only nonzero coefficient in the orthogonal list is on $p_1 = t$:

$$\frac{\langle t^3, t\rangle}{\langle t, t\rangle} = \frac{81 + 1 + 1 + 81}{9 + 1 + 1 + 9} = \frac{41}{5}.$$

Conclusion. The best approximation is $(41/5)t$. The best member of a quadratic-polynomial space need not have a nonzero quadratic term.

Exercise 6.7.11: A cubic projected onto a five-node quadratic space

Here evaluation is at $-2, -1, 0, 1, 2$, and the source's orthogonal polynomials are $p_0 = 1$, $p_1 = t$, $p_2 = t^2 - 2$. Odd-even cancellation makes the projection coefficients on p_0, p_2 zero. The coefficient on t is

$$\frac{\sum t_i^4}{\sum t_i^2} = \frac{16 + 1 + 0 + 1 + 16}{4 + 1 + 0 + 1 + 4} = \frac{17}{5}.$$

Conclusion. The orthogonal projection is $(17/5)t$.

Exercise 6.7.12: Choose a scaled orthogonal cubic

Subtract the projection from Exercise 11 to obtain $r(t) = t^3 - (17/5)t$. Its values at the five nodes are $(-6/5, 12/5, 0, -12/5, 6/5)$. Multiplying by $5/6$ yields

$$\boxed{p_3(t) = (5t^3 - 17t)/6},$$

with the requested values $(-1, 2, 0, -2, 1)$. The remainder is orthogonal to all of $\mathrm{Span}\{p_0, p_1, p_2\}$. Its degree is three, so it is nonzero and adds a new direction. Thus the four polynomials are an orthogonal basis for $\mathcal{P}_3$.

Exercise 6.7.13: Transport the Euclidean product through an invertible map

Define $\langle u, v\rangle = (Au)^T(Av)$. Symmetry follows from symmetry of the dot product. For addition, $A(u+w) = Au + Aw$, so $\langle u+w, v\rangle = (Au+Aw)^T Av = \langle u, v\rangle + \langle w, v\rangle$. Similarly $A(cu) = cAu$ gives homogeneity. Finally $\langle u, u\rangle = \|Au\|^2 \geq 0$. If it is zero, $Au = 0$; invertibility gives $u = 0$. Conversely $u = 0$ clearly gives zero. All four axioms hold. The last step is why invertibility, rather than merely linearity of A, was required.

Exercise 6.7.14: The same construction for an injective transformation

Define $\langle u, v\rangle = T(u) \cdot T(v)$. Symmetry and the two linearity axioms follow exactly by applying linearity of T and the dot-product rules in $\mathbb{R}^n$. The self-product is $\|T(u)\|^2$, nonnegative and zero exactly when $T(u) = 0$. Because T is one-to-one and $T(0) = 0$, this happens only for $u = 0$. Thus positive definiteness holds, proving an inner product on V. Without injectivity, a nonzero kernel vector would have self-product zero and violate that axiom.

Exercise 6.7.15: Move a scalar out of the second argument

Symmetry and homogeneity in the first argument give

$$\langle u, cv\rangle = \langle cv, u\rangle = c\langle v, u\rangle = c\langle u, v\rangle.$$

These are real inner products, so no complex-conjugation convention is involved.

Exercise 6.7.16: The distance between two orthonormal vectors

Expand using bilinearity and symmetry:

$$\|u - v\|^2 = \langle u, u\rangle - 2\langle u, v\rangle + \langle v, v\rangle = 1 - 0 + 1 = 2.$$

Taking the nonnegative square root gives $\boxed{\|u - v\| = \sqrt{2}}$.

Exercise 6.7.17: Recover an inner product from two norms

Expand the two squared norms: $\|u+v\|^2 = \|u\|^2 + 2\langle u, v\rangle + \|v\|^2$ and $\|u-v\|^2 = \|u\|^2 - 2\langle u, v\rangle + \|v\|^2$. Subtracting cancels both self-products and leaves $\|u+v\|^2 - \|u-v\|^2 = 4\langle u, v\rangle$. Division by four proves the stated identity.

Exercise 6.7.18: The parallelogram identity

Add, instead of subtracting, the two expansions in Exercise 17. The mixed terms $2\langle u, v\rangle$ and $-2\langle u, v\rangle$ cancel, leaving

$$\boxed{\|u+v\|^2 + \|u-v\|^2 = 2\|u\|^2 + 2\|v\|^2}.$$

The proof uses only inner-product axioms, so it is not restricted to coordinate vectors.

Exercise 6.7.19: Arithmetic mean dominates geometric mean

Use the ordinary dot product on the two supplied coordinate vectors. Their dot product is $2\sqrt{ab}$, while each squared norm is $a+b$. Cauchy–Schwarz gives $2\sqrt{ab} \le \sqrt{a+b}\sqrt{a+b} = a+b$, where nonnegativity of a, b makes the square roots defined and the left side nonnegative. Divide by two: $\boxed{\sqrt{ab} \le (a+b)/2}$. The case $a = b = 0$ is included.

Exercise 6.7.20: The square of a mean versus the mean of squares

For $u = (a, b)^T$ and $v = (1, 1)^T$, squared Cauchy–Schwarz gives $(a+b)^2 \le (a^2+b^2)(1+1) = 2(a^2+b^2)$. Divide by four to obtain

$$\boxed{\left(\frac{a+b}{2}\right)^2 \le \frac{a^2+b^2}{2}}.$$

This holds for all real a, b, with no nonnegativity assumption needed.

Exercise 6.7.21: An integral inner product that vanishes

The product expands as $(1-3t^2)(t-t^3) = t - 4t^3 + 3t^5$. Therefore

$$\langle f, g\rangle = \int_0^1 (t - 4t^3 + 3t^5)\,dt = \frac{1}{2} - 1 + \frac{1}{2} = \boxed{0}.$$

Both functions are continuous, so this is the specified inner product; they are orthogonal even though neither is zero.

Exercise 6.7.22: Another orthogonal pair of functions

Expanding $(5t-3)(t^3-t^2)$ gives $5t^4 - 8t^3 + 3t^2$. Thus $\langle f, g\rangle = \int_0^1 (5t^4 - 8t^3 + 3t^2)\,dt = 1 - 2 + 1 = \boxed{0}$. The integral of a product can cancel even when the functions have nonzero norms.

Exercise 6.7.23: An integral norm is a square root

For $f(t) = 1 - 3t^2$,

$$\|f\|^2 = \int_0^1 (1 - 6t^2 + 9t^4)\,dt = 1 - 2 + \frac{9}{5} = \frac{4}{5}.$$

Hence $\boxed{\|f\| = 2/\sqrt{5}}$. Nonnegativity determines the choice of square root.

Exercise 6.7.24: The norm of a cubic-minus-quadratic function

For $g(t) = t^3 - t^2$,

$$\|g\|^2 = \int_0^1 (t^6 - 2t^5 + t^4)\,dt = \frac{1}{7} - \frac{1}{3} + \frac{1}{5} = \frac{1}{105}.$$

Thus $\boxed{\|g\| = 1/\sqrt{105}}$.

Exercise 6.7.25: An orthogonal polynomial basis on a symmetric interval

On $[-1,1]$, oddness gives $\langle 1,t\rangle = 0$ and $\langle t,t^2\rangle = 0$. The projection of t^2 onto 1 has coefficient $(\int_{-1}^{1} t^2)/(\int_{-1}^{1} 1) = (2/3)/2 = 1/3$. Thus $t^2 - 1/3$ is orthogonal to both lower-degree polynomials. A conventional scaling gives the basis

$$\boxed{\{1,\ t,\ (3t^2-1)/2\}}.$$

The last polynomial has integral zero and its product with t is odd. All three are nonzero of distinct degrees, so they form a basis of $\mathrm{Span}\{1,t,t^2\}$. Their squared norms are $2, 2/3, 2/5$, respectively.

Exercise 6.7.26: Repeat the orthogonalization on a wider interval

On $[-2,2]$, the first two polynomials are again orthogonal by oddness. The coefficient to subtract from t^2 is $(\int_{-2}^{2} t^2)/(\int_{-2}^{2} 1) = (16/3)/4 = 4/3$. Scaling the remainder by 3 gives $\boxed{\{1,t,3t^2-4\}}$. The last polynomial has integral $3(16/3) - 4(4) = 0$, and its product with t is odd. Distinct degrees give independence and the same polynomial span. All squared norms are positive; the last is $256/5$.

Exercise 6.7.27: Complete the five-node orthogonal polynomial basis

Use the evaluation nodes $-2,-1,0,1,2$ and start with $p_0 = 1, p_1 = t, p_2 = t^2 - 2$. They are orthogonal. For t^3, only the projection onto t is nonzero, with coefficient $17/5$ as computed in Exercise 11. Set $r_3 = t^3 - (17/5)t$. For t^4, oddness makes the projections onto p_1 and r_3 zero. The constant projection coefficient is $34/5$, while

$$\frac{\langle t^4, p_2\rangle}{\langle p_2, p_2\rangle} = \frac{(\sum t_i^6) - 2(\sum t_i^4)}{14} = \frac{130-68}{14} = \frac{31}{7}.$$

Hence the final remainder is

$$r_4 = t^4 - \frac{34}{5} - \frac{31}{7}(t^2-2) = t^4 - \frac{31}{7}t^2 + \frac{72}{35}.$$

One convenient rescaled basis is

$$\boxed{\{1,t,t^2-2,5t^3-17t,35t^4-155t^2+72\}}.$$

Their distinct degrees prove independence, and Gram–Schmidt preserves the original span $\mathcal{P}_4$. Exact dot products of their five-node value vectors are zero for every distinct pair, verifying orthogonality.

Exercise 6.7.28: Orthogonalize powers of cosine by integration

Start with $q_0 = 1$ and $q_1 = \cos t$: their integral product over $[0, 2\pi]$ is zero. Since $\int \cos^2 t = \pi$ and $\int 1 = 2\pi$, the projection of $\cos^2 t$ onto the constant is $1/2$. Its projection onto $\cos t$ is zero because $\int_0^{2\pi} \cos^3 t\, dt = 0$. Thus $q_2 = \cos^2 t - 1/2$. For $\cos^3 t$, the constant and q_2 projections vanish by cancellation over a half-period. The remaining coefficient is

$$\frac{\int_0^{2\pi} \cos^4 t\, dt}{\int_0^{2\pi} \cos^2 t\, dt} = \frac{3\pi/4}{\pi} = \frac{3}{4}.$$

Therefore $q_3 = \cos^3 t - (3/4)\cos t$, giving the orthogonal basis

$$\boxed{\{1, \cos t, \cos^2 t - 1/2, \cos^3 t - (3/4)\cos t\}}.$$

The squared norms are $2\pi, \pi, \pi/4, \pi/16$, all nonzero. Equivalently the last two are $(1/2)\cos 2t$ and $(1/4)\cos 3t$, so the basis spans the same four-dimensional space as the original list.